LESSON 6.4

Write and Evaluate Numerical Expressions with Exponents

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

WHY THIS MATTERS The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

REMEMBER

Write and Evaluate Numerical Expressions with Exponents

Order of Operations (PEMDAS) A numerical expression is a math phrase made of numbers and operations, with no equal sign. Writing one for a whole situation keeps every part visible; evaluating it means finding its single value, following the order of operations.

1. Grouping Symbols: Parentheses, brackets, fraction bars inner to outer.
2. Exponents: Evaluate all powers.
3. Multiply & Divide left to right; Add & Subtract left to right

Words: **Numerical expression** ***(Expresión numérica)*** **Evaluate** ***(Evaluar)*** **Order of operations** ***(Orden de las operaciones)*** **Power** ***(Potencia)*** **Base** ***(Base)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression shows 5^3 as repeated multiplication?

(A) $3 \times 3 \times 3 \times 3 \times 3$

(B) $5 \times 5 \times 5$

(C) 5×3

(D) $5 + 5 + 5$

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A rideshare app charges $2.50 plus $1.75 per mile. The expression for the fare is 2.50 + 1.75m, where m is miles traveled. Your friend lives 8 miles away. What does the 2.50 represent?

(A) The cost of one mile
(B) A flat base fee charged no matter how far you go
(C) The number of miles
(D) The tip

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the value of 6^2?

(A) 8
(B) 12
(C) 36
(D) 62

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the value of 4^3?

(A) 7
(B) 12
(C) 43
(D) 64

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

In 2^5, which number is the base and which is the exponent?

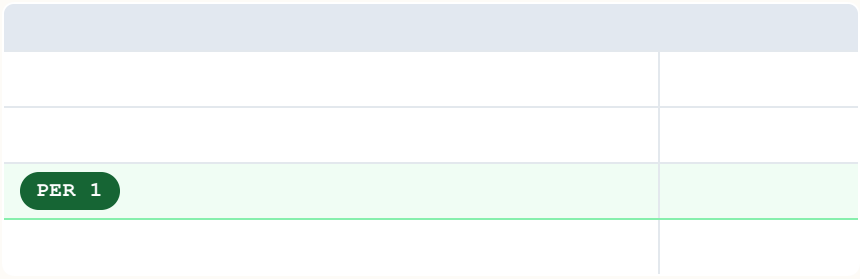
(A) Base 2, exponent 5
(B) Base 2, exponent 2
(C) Base 5, exponent 2
(D) Base 5, exponent 5

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A rideshare app charges $2.50 plus $1.75 per mile. The expression for the fare is 2.50 + 1.75m, where m is miles traveled. Your friend lives 8 miles away. What is 2.50 + 1.75(8)?

(A) 4.25
(B) 14.00
(C) 16.50
(D) 34.00



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

EXPLAIN YOUR THINKING

Two students evaluate 3 + 4 × 2 and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?

SENTENCE STARTER The correct value is ___ because ___ comes before ___. Without an agreed order, the same expression would have ___ values.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.4

Write and Evaluate Numerical Expressions with Exponents

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

WHY THIS MATTERS The studio's booking sheet lists a price for each kind of seat. Writing one numerical expression for the whole order — and evaluating it in the right order — is faster and easier to check than adding costs one at a time. Some counts are written as powers, so the exponent has to be handled before any multiplying. The booking sheet lists a $150 base fee plus $12 for each speaker rented. Write one expression for the cost of renting 5 speakers.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A rideshare app charges $2.50 plus $1.75 per mile. The expression for the fare is 2.50 + 1.75m, where m is miles traveled. Your friend lives 8 miles away. What does the 1.75 represent?

(A) The base fee
(B) The total cost
(C) The number of miles
(D) The cost per mile, multiplied by the number of miles

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each expression by which operation you do FIRST.

Groups: Do the power first | Do the parentheses first | Do the multiplication first

_____ $5 + 2^3$

_____ $(5 + 2) \times 3$

_____ $5 + 2 \times 3$

_____ $10 - 3^2$

_____ $(10 - 3) \times 2$

_____ $10 - 3 \times 2$

3 **Write the letter of the matching item on each line.**

Match each power to its value.

_____ 2^3	(A) 8
_____ 3^2	(B) 9
_____ 5^2	(C) 25
_____ 10^3	(D) 1,000

PART 2 Practice the skill

show every step

4 **Circle the letter of the best answer. Show how you know.**

Evaluate 6 + 2 × 5.

(A) 13 (B) 16

(C) 30 (D) 40

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

PART 3 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Evaluate 4^2.

(A) 6 (B) 8

(C) 16 (D) 44

Rewrite the whole expression each line	What I simplified

6 **Circle the letter of the best answer. Show how you know.**

Which numerical expression includes an operation and has no variable or equal sign?

(A) $x + 5$

(B) $8 \times 3 + 2$

(C) $8 \times 3 = 24$

(D) 24

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Two students evaluate $3 + 4 \times 2$ and get 14 and 11. Which one followed the order of operations, and why does the order have to be agreed on at all?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help